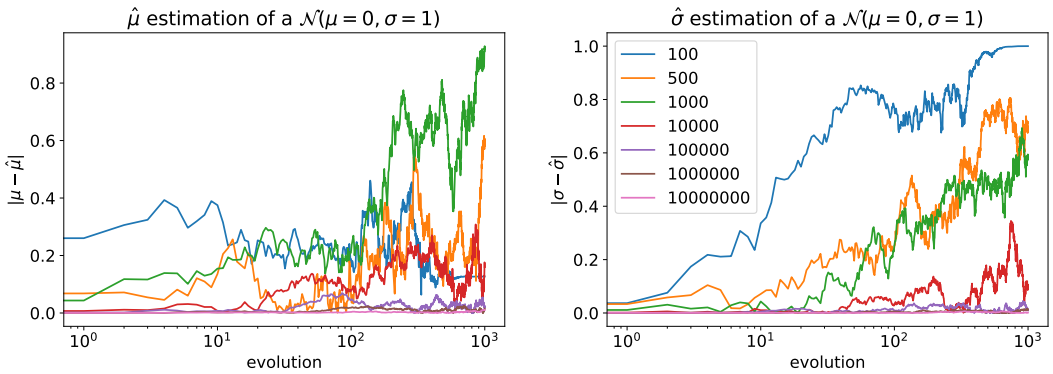
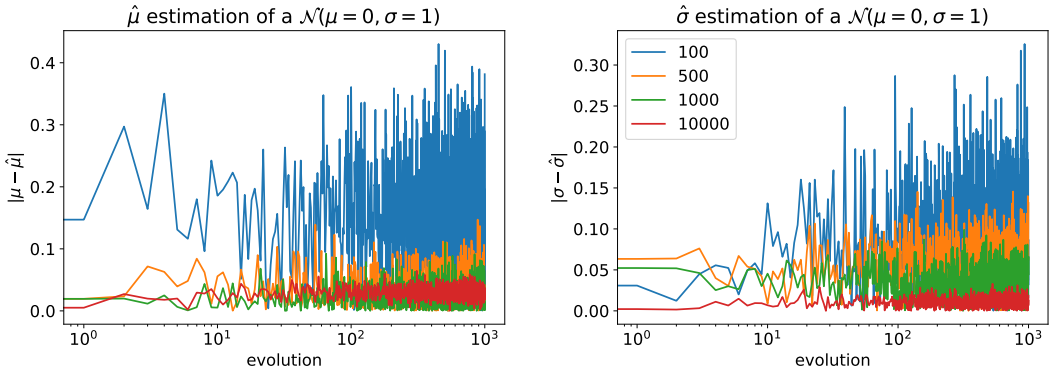


(a) Mean estimation

(b) Standard Deviation

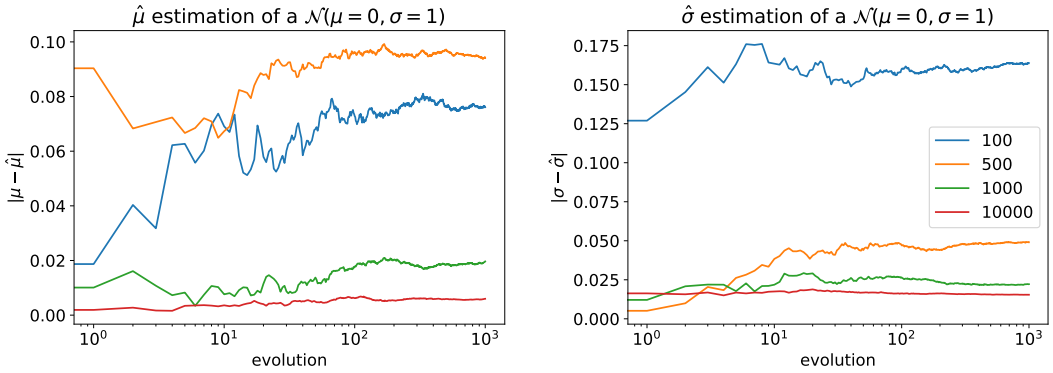
Figure 4: Recursive fitting-sampling of a 1D Gaussian with different numbers of samples drawn. We find that unless sampled a very large number of times, *i.e.* <100000, both standard deviation and mean get significantly affected. Here we report a single run; while re-running the experiment changes the initial performance, both μ and σ drift over time. The overall graph looks quite similar to that of a Gaussian random walk.



(a) Mean estimation

(b) Standard Deviation

Figure 5: Recursive fitting-sampling of a 1D Gaussian with different numbers of samples drawn. In this plot data get accumulated in a pool, from which a fixed sample is drawn. In other words, a model n gets data sampled, its output is mixed with data sampled from models $1 \ldots n$, and then the mix gets sampled to fit the model $n + 1$. The uncertainty arising from all of the different modalities appearing in data causes the distribution parameters to jump around quite significantly.



(a) Mean estimation

(b) Standard Deviation

Figure 6: Recursive fitting-sampling of a 1D Gaussian with different number of samples drawn. In this plot data are accumulated in a pool, all of which is used to fit a model. In other words, a model n gets data sampled, its output mixed with data sampled from models $1 \ldots n$, and then the result is used to fit the model $n + 1$. Over time the variance in estimates reduces due to linear growth of data.

This means that as $n \to \infty$, the variance diverges linearly. This is the same scaling as for a single dimensional Gaussian random walk. We can further see the similarities in numerical experiments shown on Figure 4 for a range of different sample sizes, confirming these theoretical intuitions.

Even though the variance of X_j^n diverges, it does not provide us with any information of what the corresponding estimates of μ_{n+1} and σ_{n+1}^2 are, or how far they are from the original μ and σ. In particular, we may want to consider what the distance would be between the true distribution and the approximated distribution at step $n+1$. To measure this we can consider the Wasserstein-2 distance between two normals:

$$R_{W_2}^{n+1} := W_2^2\left(\mathcal{N}(\mu, \sigma^2), \mathcal{N}(\mu_{n+1}, \sigma_{n+1}^2)\right) = \|\mu_{n+1} - \mu\|^2 + \|\sigma_{n+1} - \sigma\|^2$$

Now we can calculate the risk that occurs due to finite sampling, *i.e.* what the expected value of the distance is (expanding in $1/M_i$):

$$\mathbb{E}_{\mu_{n+1}, \sigma_{n+1}^2}\left[R_{W_2}^{n+1}\right] = \frac{3}{2}\sigma^2\left(\frac{1}{M_0} + \frac{1}{M_1} + \cdots + \frac{1}{M_n}\right) + \mathcal{O}(2), \tag{4}$$

$$\mathrm{Var}_{\mu_{n+1}, \sigma_{n+1}^2}\left[R_{W_2}^{n+1}\right] = \frac{1}{2}\sigma^4\left(\frac{3}{M_0^2} + \frac{3}{M_1^2} + \cdots + \frac{3}{M_n^2} + \sum_{i \neq j}\frac{4}{M_i M_j}\right) + \mathcal{O}(3). \tag{5}$$

This result allows us to interpret exactly what occurs in this formulation of *model collapse*. To be precise, due to errors occurring from re-sampling the approximated distribution, each generation ends up corresponding to a new step in a random walk of model parameters. The risk that occurs in this model ends up diverging for a constant sample size at each generation. In order for the end distribution approximation to be accurate, and for the distance to be finite, the sampling rate M_i needs to increase superlinearly, *i.e.* one needs to collect increasingly more samples over time, perhaps quadratically. However, even in that case the expected distance after n steps remains non-zero and the only case in which it does in fact end up being 0 is when sampling is infinite at each step. Overall, this only shows us how far on average we go from the original distribution, but the process can only 'terminate' if the estimated variance at a certain generation becomes small enough, *i.e.* we effectively turn into a delta function.

Shown on Figures 5 and 6 are different runs of this process for different values of parameters of $\alpha_i, \beta_i, \gamma_i$ for different sample sizes, which was investigated numerically to see whether they can be enough to overcome *model collapse*, however even in those cases the changes are inevitable, although attenuated.

4.3 Noisy approximation model

With the simple example out of the way, we can now construct a lower bound on the distance of generation n distribution from the original and show that without superlinear sampling it similarly diverges in the limit of large n. A nice property of Wasserstein-2 distance is that Gaussians provide a universal lower bound for the Wasserstein distance [Gelbrich, 1990]. In particular, for κ and ν probability measures on a Euclidean N-dimensional space with μ_κ and μ_ν means, Σ_κ and Σ_ν covariance matrices, we have that

$$W_2^2(\kappa, \nu) \geq \|\mu_\kappa - \mu_\nu\|^2 + \mathrm{Tr}\left(\Sigma_\kappa + \Sigma_v - 2\left(\Sigma_\kappa^{1/2}\Sigma_v\Sigma_\kappa^{1/2}\right)^{1/2}\right) \geq \|\mu_\kappa - \mu_\nu\|^2$$

With this, instead of quantifying the distance exactly, we can instead lower bound it. The only limitation is that we are going to have to specify a functional approximation model. In order to achieve a W_2 bound, we will be required to specify how the mean changes between generations. In the scenario where we only have access to the sample mean, we would approximate the mean of the next generation distribution as Equation (1). However, as more information arrives, or the model begins using it better, we may end up diverging from the sample mean. We would still require that the model have good performance, *i.e.* on average the mean estimate is the same. We will also have to specify expected behaviour of the model over the the variance calculation, which once again will be chosen such that it averages out. Thus, we will adopt the following evolution over generations:

$$\mu_{i+1} = \frac{1}{M_i}\sum_j X_j^i + \varepsilon_{i+1} = \frac{\Sigma_i^{1/2}}{\sqrt{M_i}}T^{i+1} + \mu_i + \varepsilon_{i+1}; \quad \mathbb{E}_{X_j^i}(\Sigma_{i+1}) = \Sigma_i \tag{6}$$

where we define T^{i+1} to satisfy the equation above, *i.e.* $T^{i+1} = \frac{\Sigma_i^{-1/2}}{\sqrt{M_i}}\sum_j\left(X_j^i - \mu_i\right)$. With this normalisation T has mean 0 and covariance I_N and by the central limit theorem (CLT) we would have $T^{i+1}|\mu_i, \Sigma_i \xrightarrow{\mathcal{D}} \mathcal{N}(0, I_N)$, however the lower bound will not rely on this. To arrive at a lower bound for the risk, similar to that of Equation (4), we are going to have to make a few assumptions about the form of ε_{i+1}.

Assumptions:

1. On average we can capture the mean to be the same as at the iteration prior:
$$\mathbb{E}[\varepsilon_{i+1}|\mu_i, \Sigma_i] = 0 \tag{7}$$
2. Given all of X_j^i, epsilon must be constant, *i.e.* it is a function of only the data:
$$\varepsilon_{i+1} = \varepsilon_{i+1}\left(X_j^i\right) \tag{8}$$
In particular, it is dependent on μ_i and Σ_i only through the data.
3. The extra noise is orthogonal to the sample mean in the sense of random variables. This is effectively assuming that the noise does not contain any first moment information, *i.e.* we have:
$$\mathrm{Cov}(\varepsilon_{i+1}, T^{i+1}|\mu_i, \Sigma_i) = 0 \tag{9}$$
This may seem like a rather strong assumption, compared to the previous ones, however this property can be shown to hold true when imposing CLT on T^{i+1} in the limit of large M_i (note here that M_i can only be assumed to be **large**, and not infinite) and assuming that ε is strictly a function of moments higher than first. Furthermore, a property of this type is necessary to actually provide any information, since prior to it there would be no need to separate into an epsilon term and a sample mean term, since all could be absorbed into ε.

In Appendix A.2, we further provide an alternative to Assumption 3, wherein by bounding the size of noise we are able to recover a similar bound, but only as an expansion in $1/M_i$.
With all the assumptions in place, we now have the following bound:

$$\mathbb{E}\left[R_{W_2}^{i+1}\right] \geq \mathbb{E}\left(\|\mu_{i+1} - \mu\|^2\right) \tag{10}$$
$$= \mathbb{E}\left(\|\mu_i - \mu\|^2\right) + \mathbb{E}\left(\|\varepsilon_{i+1}\|^2\right) + \frac{1}{M_i}\mathbb{E}\left((T^{i+1})^\top \Sigma_i (T^{i+1})\right) + \tag{11}$$
$$+ \frac{2}{\sqrt{M_i}}\mathbb{E}\left((\varepsilon_{i+1})^\top \Sigma_i^{1/2} T^{i+1} + (\mu_i - \mu)^\top \Sigma_i^{1/2} T^{i+1}\right) \tag{12}$$
$$= \mathbb{E}\left(\|\mu_i - \mu\|^2\right) + \frac{\mathrm{Tr}\,\Sigma}{M_i} + \mathbb{E}\left(\|\varepsilon_{i+1}\|^2\right) + \frac{2}{\sqrt{M_i}}\mathbb{E}\left((\varepsilon_{i+1})^\top \Sigma_i^{1/2} T^{i+1}\right) \tag{13}$$

Now the only quantity to evaluate is
$$\frac{2}{\sqrt{M_i}}\mathbb{E}\left((\varepsilon_{i+1})^\top \Sigma_i^{1/2} T^{i+1}\right) = \frac{2}{\sqrt{M_i}}\int d\Sigma_i\, p(\Sigma_i)\,\mathrm{Tr}\left[\Sigma_i^{1/2}\,\mathrm{Cov}(\varepsilon_{i+1}, T^{i+1}|\Sigma_i)\right] = 0, \tag{14}$$
by Assumption 3. Therefore, the overall bound would be similar to the Gaussian case, but with extra noise variance terms:
$$\mathbb{E}_{\mu_{n+1},\sigma_{n+1}^2}\left[R_{W_2}^{n+1}\right] \geq \mathrm{Tr}\,\Sigma\left(\frac{1}{M_0} + \frac{1}{M_1} + \cdots + \frac{1}{M_n}\right) + \sum_{i=1}^{n+1}\mathbb{E}\left(\|\varepsilon_i\|^2\right) \tag{15}$$

As a result, we have shown that the same superlinear scaling would be required to minimise the lower bound on *model collapse* even in the case of more generic models of approximation, in which the mean at step $i+1$ can be separated orthogonally into the sample mean and 'extra'.

Overall, the message of this section can be summarised as follows:

When learning on generational data, due to finite sampling, we are only able to ***approximate*** *the original distribution. While on average we should recover the original distribution, the variance arising from this is non-zero. As a result, over the generations, the average distance of n'th generation from the original grows and can become infinite in the limit since errors compound over time.*

5 Evaluation

5.1 Training from scratch with GMMs and VAEs

Gaussian Mixture Models. In this subsection we evaluate the performance of Gaussian Mixture Models (GMM) [Reynolds et al., 2009]. The underlying task here is that a given GMM tries to separate two artificially-generated Gaussians. Figure 7 shows the progression of the GMM fitting process over time. The left-most plot shows the original two Gaussians with the ground truth labels. The next plot shows the GMM fitted on the original data with no cross-generational data used *i.e.* $\alpha_i = \gamma_i = 0$, where the error is minimal. Yet, within 50 iterations of re-sampling we arrive to a point where the underlying distribution is mis-perceived. The performance worsens over time and by iteration 2000 we arrive at a point estimate of the distribution with very little variance. The L2 distance between the original GMM and its descendants is plotted in Figure 13.